WILSON

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Principal Engineer — Payment & Transaction-Processing Platforms | Hands-On Java Architecture & Resilience

PROFESSIONAL SUMMARY

Hands-on technical lead for mission-critical payment platforms, with 18+ years across Tier-1 banking, fintech and SaaS products, and consumer e-commerce, combining Java engineering with solution architecture. Works across the full lifecycle of transaction-processing systems — architecture and solution design, core code and code review, migration of live systems and production resilience.

Since 2022, a Vice President in UOB's Group Technology & Operations, leading 25 onshore and offshore engineers and chairing governance of four IT service vendors for cross-border payment and retail-lending platforms that serve millions of accounts in a MAS-regulated bank. Owns the architecture of the bank's payment-initiation hub and led its move to ISO 20022 messaging.

Began as an engineer and technical lead on Renren's e-commerce platform, Sohu's news CMS and Baidu's Nuomi merchant system, then spent 2016–2022 in head-of-technology and architect roles at start-ups and scale-ups, delivering microservices platforms, a payment subsystem and a multi-tenant SaaS platform from scratch. At UOB, reports status, risks and decisions to senior management, the steering committee and the business team.

CORE COMPETENCIES

Payment Systems	Payment initiation, validation, enrichment and status tracking; hand-off to core banking for settlement; reconciliation; ISO 20022, SWIFT; card and e-wallet payment integration
Architecture & Integration	Java, Spring Boot, Apache Camel (enterprise integration patterns), API design, event-driven integration, microservices, distributed systems, multi-tenant SaaS
Performance & Resilience	Low-latency transaction processing, concurrency control, thread-pool and JVM tuning, high availability, disaster recovery (DR) and failover drills, capacity baselines
Security & Risk	API and database encryption, credential management, role-based access control, secure SDLC, data protection impact assessments (DPIA), data governance, MAS TRM (Singapore technology-risk rules)
Engineering Delivery	DevSecOps practice: CI/CD on Bitbucket and Jenkins, SonarQube quality gates, automated regression and penetration testing; Splunk observability; SIT/UAT and Change Advisory Board (CAB) release approval
Platforms & Middleware	Linux (RHEL); MySQL, SQL, Redis; Kafka, RocketMQ and AWS SQS/SNS messaging; Docker, Kubernetes; AWS (API Gateway, EC2/ECS, RDS, CloudWatch)
Technical Leadership	Design and code reviews, quality gates, engineering standards (contributed bank-wide), mentoring, 1:1s and performance management, steering-committee reporting

SELECTED ACHIEVEMENTS

- **Hands-on core engineering:** Personally built the hub's core modules — payment-transaction initiation, concurrency control, the corporate- and user-role permission matrix and processing of payment status results returned by downstream systems.
- **Payment-hub design:** Led end-to-end design of the hub — API contracts, orchestration, validation, enrichment and status tracking — integrating digital and branch channels, reference data, and vendor and third-party banking systems.
- **Zero-interruption modernisation:** Planned and architected the phased move of cross-border payments to ISO 20022 (SWIFT's MT-to-MX migration), running old and new message formats side by side: new-format messaging went live in May and August 2025, ahead of SWIFT's November 2025 deadline, and the old path was retired in August 2026 with no service interruption.
- **Performance engineering:** Reduced end-to-end transaction latency under high load through thread-pool and JVM tuning and removal of synchronous blocking; earlier brought VV's services to millisecond-level response times.
- **Resilience and security:** Designed security controls — API and database encryption of sensitive data and credential management — for payment, lending and customer-data platforms under secure SDLC, DPIA and MAS TRM; personally led DR and failover drills for the real-time payment middleware, sustaining 99.9%+ availability under peak load.
- **Payment integrations across markets:** Designed and coded Renren Mall's payment integration (Alipay, UnionPay) and Baidu Nuomi's payment-gateway integration (Alipay, WeChat Pay, UnionPay); architected VV-Life's payment subsystem, integrating UnionPay, Visa/Mastercard, Google Pay and GrabPay payments with settlement reconciliation.
- **Engineering governance:** Established delivery KPIs and CI/CD quality gates — 99% SonarQube coverage on all code, automated regression and penetration testing, SIT/UAT sign-off for every release — and contributed to bank-wide SDLC and code-review standards.

PROFESSIONAL EXPERIENCE

United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations

Two mission-critical platform lines: cross-border payments on SWIFT ISO 20022 and retail unsecured lending and collections.

Architecture & Hands-On Engineering

- Stay hands-on alongside delivery: review newly submitted core pull requests daily, co-design solutions with engineers, own the architecture, write core modules for significant changes and resolve production issues.
- Lead the 10-engineer backend team of a regional customer-360 platform live since 2022: evaluated Cloudera as its big-data processing framework, designed the customer data models and built its data-analysis module and Spring Batch jobs.
- The payment-initiation hub hands validated payments to core banking, which executes settlement, and processes the results returned; the customer-360 platform takes daily account summaries from core banking for portfolio analysis.
- Designed and built a production AI code-review service: headless Claude Code agents, orchestrated by Spring Boot, review each pull request alongside SonarQube. Live since May 2026 after formal architecture and risk approval, it cut senior engineers' review time by an estimated ~80%; the AI never approves or merges.

Delivery, Production & Leadership

- Led a 15-engineer onshore/offshore team and owned end-to-end delivery of the ISO 20022 programme — scope, roadmap, risk and issue tracking, release planning and production rollout — shipping upgrades every 2–3 months after go-live.
- Personally prepare the change request for every production release on both platform lines and present the go-live plan, risk controls, rollback plan, test results and database changes to the Change Advisory Board (CAB).
- Coordinate P1–P3 production incidents end to end — escalation, stakeholder communication, root-cause analysis (RCA) and post-incident reports — and produce technical incident analyses that support the bank's regulatory reporting to MAS.
- Own production support for the payment platforms, with no separate site reliability engineering (SRE) team: SLA tracking, service-level indicators/objectives (SLIs/SLOs), Splunk monitoring and alerting, and capacity baselines.
- Resolve schedule conflicts between the two platform lines by agreeing MVP scope with the requirement, project and technical leads, confirming dependencies and setting a sequence all parties accept.
- Own technology for retail unsecured-lending servicing — payment initiation, validation and lifecycle tracking — and post-disbursement collections across credit-card and unsecured-loan portfolios.
- Lead both platform teams through squad leads and engineering managers — 1:1s, mentoring and performance management — and work closely with teams in India, Thailand and Indonesia across time zones.
- Hold decision or joint-decision authority on technology investment, vendor selection and statement-of-work (SOW) and licence negotiation, and run vendors' periodic budget reviews, year-end performance reviews and accountability meetings.

Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

Project Manager & Architect

Multi-tenant SaaS for online-game businesses, built 0-to-1 on AWS — KYC, bonus incentives, VIP tiers and multi-channel payments.

- Led a 40-person project team, including outsourced staff, introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness.
- Architected and delivered the platform on Spring Cloud and Kubernetes behind an API gateway, including hierarchical KYC identity verification and rule models for authentication levels and bonus eligibility, triggers and distribution.
- Architected the AWS deployment (API Gateway, EC2/ECS, RDS, DynamoDB) and applied DPIA, API and database encryption and credential management; led data migration.

VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

Head of Technical Centre

VV-Life: a local-life services platform connecting offline merchants — food, fresh produce, travel and everyday services.

- Led VV's 140-person technology organisation through its managers and built its microservices platform from scratch, owning technology selection and the technology budget.
- Took part in product-strategy planning and delivered the first VV-Life release in 2019, including the payment subsystem's transaction, management and reconciliation modules.
- Designed the shared back-end platform architecture (Spring Boot, MySQL cluster, Redis, RocketMQ) for a B2B online-office product serving Southeast Asia, governing solution design, cost evaluation, staffing and acceptance.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

AI Park | China — Apr 2016 – Sep 2017

VP of Technical Department

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud for automated management, online payment and recommendations.

- Brought Spring Cloud microservices into the architecture, with Kafka for asynchronous flows and caching to shield the database, achieving millisecond-level service processing; led the big-data team on HBase, MongoDB and Spark Streaming.

EARLIER EXPERIENCE

- **Hengshunyuan Information Technology — Co-founder (Oct 2017 – Sep 2019, China):** Owned product, architecture and delivery of an audio-learning platform, reaching a peak of ~100k daily active users.
- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** As project leader, re-architected the merchant-side order system on Dubbo microservices and delivered two months ahead of plan.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Led the re-architecture of the fifth-generation CMS onto Spring MVC and a Redis cluster, and delivered its OpenAPI messaging platform.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM on Spring, Solr, Redis, MySQL and Nginx.

EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)